



PTH 6132: MOVEMENT SYSTEMS

Spring 2025

Class: 11:00 am -12:30 pm, T and TH in SRH 114
LAB: Sec 01 1:00- 4:00 pm, T; Sec 02 1- 4 pm, TH in SRH 215

Instructor:

Dr. Cheryl Coker

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Office hours: W 12pm – 2pm or by appt.

Course Description

Movement Systems builds on Dynamic Systems I by using the concept of movement as a complex adaptive system in the evaluation of functional and dysfunctional movement patterns, identifying causes of dysfunctional patterns, analyzing and implementing a movement program to maximize function and efficiency, and minimize risk.

Student Learning Outcomes

1. Evaluate functional and dysfunctional movement patterns. (7A, 7D19, 7D20)
2. Differentiate between age-related and pathologically caused alterations in movement. (7A, 7D20)
3. Describe normal variations of joints and muscle function when observing movement patterns. (7A, 7D19, 7D20)
4. Discuss different forms of motion analysis. (7D19, 7D20)
5. Interpret kinetic and kinematic data from motion analysis devices. (7D19, 7D20)
6. Describe components of motion, muscle function and influencing factors to analyze movement specific to sport, recreation, and activity. (7A, 7D27)
7. Analyze and implement a movement program to develop and maintain optimal movement that maximizes efficiency and minimizes risk throughout the life span. (7D27)

Course Information

Credit Hours: 4

Semester Clock/Contact Hours: 84

Weekly Clock/Contact Hours: 6

Pre Requisites: Clinical Physiology, Clinical & Functional Anatomy, Neuroscience, Dynamic Systems I, Clinical Inquiry I, Patient Client Management II, Integrated Clinical II

Co Requisites: Pathophysiology & Pharmacology I, Dynamic Systems II, Movement Systems, Patient / Client Management III, Integrated Clinical III

Resources:

Coker, C.A. (2021). *Motor learning and control for practitioners*, 5e. New York, NY: Routledge, Taylor & Francis Group.

Physio U App

On Access Physiotherapy

Bezkor, E., O'Sullivan S.B., & Schmitz T.J. (2023). *Improving functional outcomes in physical rehabilitation*, 3e. Philadelphia, PA: F. A. Davis Company.

<https://fadavispt.mhmedical.com/book.aspx?bookid=3170>

Dutton, M. (2025). *Dutton's Orthopedic: Examination, evaluation, and intervention*, 7e. New York, NY: McGraw-Hill Medical.

<https://accessphysiotherapy.mhmedical.com/book.aspx?bookid=3563>

Hall S.J. (2022). *Basic biomechanics*, 9e. New York, NY: McGraw-Hill.

<https://accessphysiotherapy.mhmedical.com/Book.aspx?bookid=3314>

Houghlum, P.A., & Bertoti, D.B. (2012). *Brunnstrom's clinical kinesiology*, 6e. New York, NY: McGraw-Hill

<https://fadavispt.mhmedical.com/Book.aspx?bookid=2148>

Kisner, C., Colby, A. & Borstad, J. (2023). *Therapeutic exercise: Foundations and techniques*, 8e. Philadelphia, PA: F.A. Davis Company.

<https://fadavispt.mhmedical.com/Book.aspx?bookid=3279>

Levangie P.K., & Norkin C.C., & Lewek M.D.(Eds.), (2019). *Joint structure & function: A comprehensive analysis*, 6e. Philadelphia, PA: F. A. Davis Company.

<https://fadavispt.mhmedical.com/content.aspx?bookid=2678§ionid=220743439>

LibGuide for Physical Therapy

The Libguide for Physical Therapy is an information sharing platform created through a collaboration between the library and the DPT faculty that presents relevant resources to support students in the Physical Therapy program. Examples of information you will find include tutorials to better understand citing sources in APA and AMA style, resources on how to avoid plagiarism, quick access to relevant databases and interlibrary loan as well as a contact link to the programs' assigned librarian. To view the DPT Program Libguide go to <https://library.plymouth.edu/physical-therapy>.

Description of teaching methods and learning experiences

Readings, discussions, videos, lab sessions, case presentations, clinical problems, outside assignments

Assignments and Activities*

Skill acquisition project**	20%
Functional movement project**	15%
Tickets to Class, Lab Assignments & homework	10%
Practical Exam	10%
Exams x 3	45%
	100%

* All scheduled assignments and values are subject to change.

** Requires participation outside of class/lab meeting hours

Student Evaluation/Grading

A	94.0 – 100
A-	90.0 – 93.9
B+	87.0 – 89.9
B	83.0 – 86.9
B-	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
F	below 73.0

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 12 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C).

Course Policies

Doctor of Physical Therapy Student Handbook: Students are reminded that more detailed information on program policies, including the appropriate and ethical use of Artificial Intelligence (AI), can be found in the Plymouth State University Doctor of Physical Therapy Student Handbook.

Professional Conduct: Students are reminded of the program expectation that they follow Professional Conduct, see the Student Handbook, Professional Conduct and Reasonable Professional Expectations for the process of handling violations, remediation and consequences. Students can access the latest version of the student handbook at <https://campus.plymouth.edu/physical-therapy/wp-content/uploads/sites/78/2019/07/DPT-StudentHandbook.pdf>

Graduate Academic Policies: The DPT Program is a selective admissions cohort-structured program. Some Graduate Academic Policies may not be relevant to the program. Students should refer to the Doctor of Physical Therapy Program Handbook for specific information about Scheduling, Capstones, and Internships. The sections on Grading System and Academic Standing Policy do apply to DPT students: <https://coursecatalog.plymouth.edu/university-policies-procedures/graduate-academic-policies/>

Cell Phones: Cell phone use (including texting) is prohibited in class. Also, please respect your classmates by turning off your phone prior to class start time. Failure to comply with both policies will result in the individual having to give their cell phone to the instructor prior to the start of class time. Phones will be returned after the class has ended. In the event of extenuating circumstances, please see the instructor.

Attendance Policy: Attendance is expected at all class sessions. Students are ultimately responsible for their attendance; however faculty will assess students Professional Expectations based, in part, on their attendance (See Student Evaluation/Grading). Students who miss class for any reason are responsible for the information and work missed. Students should speak with the professor before missing the scheduled class when able.

Late Assignments: All assignments are due at the beginning of the class and must be typed unless otherwise instructed. Any assignments turned in once instruction has started will be considered late. Late assignments may be turned in up to 24 hours after the deadline but will incur a 20% penalty. Late assignments will no longer be accepted after 24 hours and a 0 grade will be given in such cases. This includes lab assignments, skill acquisition project logs, and functional movement logs. Weekends **do count** as e-mail and Canvas function 24 hours. **EXCEPTIONS:** Tickets to class and scheduled presentations. Tickets to class are designed to prepare you for the activities in class. Failure to post them prior to the start of the assigned class will result in a 0 grade for that assignment. Failure to give an assigned presentation during the scheduled class period will also result in a 0 grade for the presentation unless prior arrangements have been made with the professor because of extenuating circumstances.

Missed Exams: Make-up examinations will be given only in specific cases of documented absences. Circumstances that may warrant an absence should be discussed with the instructor before the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Partner and Group Work Expectations: On certain assignments students may be given the option to work alone or in partnership with one or more classmates. If it is a partnership assignment, then all students in the partnership will receive one common grade for the paper. The instructor will not attempt to discern relative contributions so it is up to each partner to ensure a proper work distribution on the project. All students are expected to contribute significantly to all aspects of the development of the submitted work and it is unethical to allow a partner to “sign on” to a submission if that partner did not significantly contribute to the work. Letting a partner do all the work and expecting a grade for his or her effort is considered a violation of the Plymouth State University academic dishonesty policy. Final documents submitted should be fully read and approved by all partners.

Fair Grading Policy: Fair and equitable grading reflects values to which all members of the Plymouth State community commit themselves. Grades are used to assess the relative extent to which students achieve course objectives in all for-credit courses at PSU. Academic freedom allows instructors to determine course objectives within the bounds of established curricula, and the means by which a student’s mastery of those objectives will be evaluated; and evaluate the quality of work on individual exams or assignments Information can be found at <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Fair Grading.

Academic Dishonesty: Cheating and plagiarism are very serious forms of academic dishonesty. Any use of unauthorized assistance on exams, papers, homework assignments, or other course work constitutes cheating. Knowingly providing assistance during an exam or allowing other students to copy one's work is also a serious form of academic dishonesty. Plagiarism consists of submitting written work that has been developed wholly or partially by someone else. Submitting written work in which the ideas of others have been duplicated or even paraphrased without proper reference to the author is also a form of plagiarism. Also considered plagiarism is the acquisition of term papers or other assignments from another source and subsequent presentation of these materials as the student's own work. In addition, students may not use papers in more than one course without the permission of both instructors. Any student suspected of academic dishonesty will be reported to the Program Director immediately. A full description of academic integrity, violations and procedures when a violation to academic integrity in a course is suspected can be found at: <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Academic Integrity.

Campus Accessibility Policy: Plymouth State University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you should immediately contact Campus Accessibility Services (CAS) in Speare210 (535-3300) to determine whether you are eligible for such accommodations. Academic accommodations will only be considered for students who have registered with CAS. If you have a Letter of Accommodation for this course from CAS, please provide the instructor with that information privately so that you and the instructor can review those accommodations: <https://campus.plymouth.edu/accessibility-services/>

Diversity and Inclusivity Statement: We consider this classroom to be a place where you will be treated with respect, and we welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. See this web page for the Plymouth State University IDEA Center Inclusion, Diversity, Equity, and Access. <https://www.plymouth.edu/idea-center>

Student Support Foundation (SSF): Provides short-term emergency financial assistance and long-term student support. For more information, see <https://campus.plymouth.edu/student-support-foundation/>. Student Support Foundation also runs a food pantry, located in Belknap Hall. To learn more about SSF or access the food pantry, either via open hours or a private appointment, contact the SSF advisor, at psu-ssf@plymouth.edu.

Tentative Outline of Content

Details including specific reading requirements, assignments and activities will be posted on the course Canvas page.

Module 1: Foundational Concepts for Understanding and Improving Motor Function

- A. Communication and Instruction
- B. Practice Design and Organization
- C. Feedback and Intervention

Module II: Applications

- A. Functional Movement
- B. Balance and Postural Control
- C. Sit to Stand, Stand to Sit
- D. Locomotion
- E. Reach, Grasp and Object Manipulation